Are You Getting What You Pay For?

Trust gaps in commercial LLM APIs and the case for verifiable inference

Reference: arXiv:2504.04715

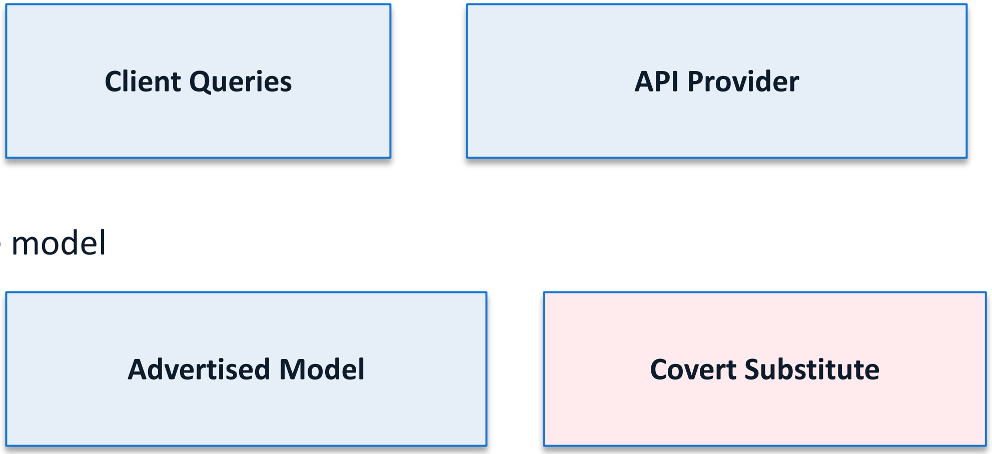
The Model Substitution Problem: Economic Incentives Drive Covert Swaps

Economic incentive: reduce inference cost while preserving perceived quality.

Substitution types observed in practice:

- Smaller model than advertised
- FP8-quantized variant of target model
- Fine-tuned variant with altered behavior
- Different model family with similar outputs

Formal test: H0 = responses from claimed model; H1 = responses from substitute model



FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Quantization Benchmark Drop (Original vs FP8)

Model Family	MMLU O	MMLU FP8	GSM8K O	GSM8K FP8	MATH O	MATH FP8	GPQA O	GPQA FP8
Llama-3.1-70B-Instruct	83.8	83.6	95.7	95.5	68.0	68.2	50.3	50.1
Qwen2.5-72B-Instruct	85.1	84.9	94.5	94.4	67.3	67.2	52.9	53.0
Mistral-Large-Instruct-2411	84.0	83.8	95.6	95.4	66.2	66.0	56.0	55.8
DeepSeek-R1-Distill-Llama-70B	82.7	82.8	95.1	95.0	69.3	69.1	49.8	49.6
Llama-3.3-70B-Instruct	86.0	85.7	96.1	95.9	70.5	70.2	54.2	53.9

Result: FP8 scores remain close to original across benchmarks, often within noise/error bars.

Higher Temperature Amplifies Performance Gaps but Increases Noise

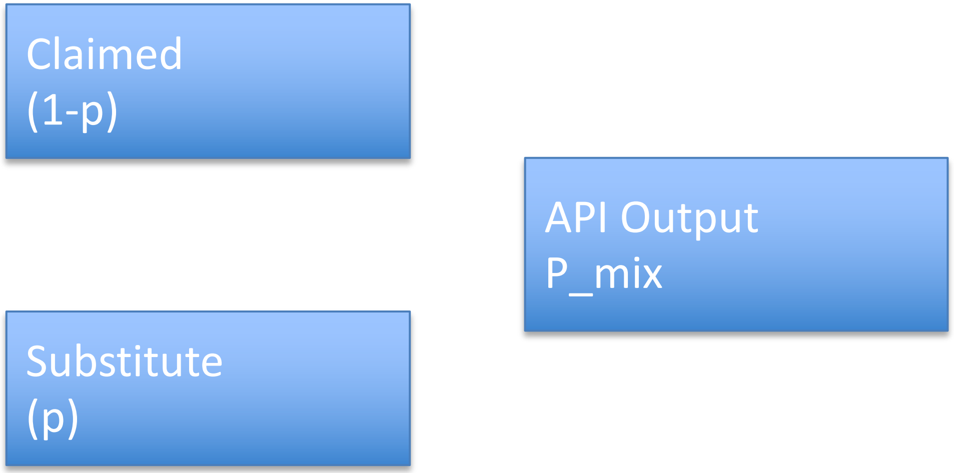
Figure 1: benchmark accuracy vs temperature for five moc
At T=2.0, absolute performance degrades broadly.
Original–substitute gaps become noisier and less stable.
Increased sampling variance weakens statistical confidenc
Detection becomes harder precisely where divergence mi

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Figure 5: identical Llama-3-8B weights, divergent log-probab
Variation appears across transformers/vLLM versions.
Variation also appears across A100 vs H100 hardware.
Hence, log-prob signatures are not stable identifiers.
Production nondeterminism defeats metadata-only auditing

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

Attack: provider routes each query to substitute model with probability p.
Mixed distribution: P_mix = (1-p)·P_claimed + p·P_substitute
As p → 0, P_mix converges to claimed-model distribution.
Statistical distinguishability shrinks; sample complexity grows rapidly.
Low-rate routing makes practical audits query-prohibitive.



Software-Only Auditing Methods Are Fundamentally Unreliable

Software-Only Auditing Methods

Method Type	Weakness	Practical Limitation
Text-based statistical tests (MMD, etc.)	Low statistical power for subtle substitutions	Requires impractically high query counts; fails under randomized routing
Log-probability fingerprinting	Inference nondeterminism across software/hardware stacks	Same model yields divergent log-probs; unstable production signatures
Adversarial countermeasures	Provider can evade fixed probes and benchmark leakage	Adaptive attacks and benchmark overfitting invalidate audit assumptions

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

TEEs shift verification from statistics to cryptographic attestation.

- Attestation binds execution to measured binaries, weights, and hardware state.
- Guarantee: declared model artifact is what actually runs at inference time.
- Compared with software tests: provable integrity, robust to adaptive adversaries.
- Operationally viable: NVIDIA confidential computing enables practical deployment.
- Modest overhead is acceptable relative to integrity and trust gains.

Key Takeaways: Close the Verification Gap with Hardware Attestation

- **Model substitution is economically rational and hard to observe externally.**
- FP8 quantization preserves benchmark quality, masking covert swaps.
- Inference nondeterminism invalidates stable log-probability fingerprints.
- Randomized substitution makes statistical auditing query-prohibitive.
- TEEs are the most actionable path to verifiable LLM API integrity.
- Call to action: standardize hardware attestation in API trust frameworks.